



SUJET STATAPP 316 - 2026-2027



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Organisme : Ekimetrics

Catégorie : Prédiction

Sujet :

Joint Uncertainty for Multivariate Electricity-Demand Forecasts

Description du sujet :

An electricity system must continuously balance electricity consumed and electricity supplied. If available production becomes insufficient relative to demand, system operators must rapidly activate reserves, import electricity or, in extreme situations, disconnect part of the load; sufficiently severe imbalances can contribute to blackouts. Excess production must similarly be absorbed, exported or curtailed.

Because large-scale electricity storage remains limited relative to total grid demand, these adjustments cannot rely on storing large quantities of electricity until they are needed. Operators therefore need to anticipate demand in order to schedule production and reserves in advance.

Day-ahead forecasting consequently concerns not one quantity but a complete "demand trajectory": $Y=(Y_1, \dots, Y_{24})$, where Y_h represents electricity consumption during hour h of the following day.

Students will first build a reasonable day-ahead forecasting model from historical data. This forecasting step is necessary to obtain predictions to analyse, but it is "not the main methodological focus of the project". The central question concerns the uncertainty around the resulting 24-dimensional forecast.

A standard approach would be to construct a separate 90% prediction interval for each hour. This can be misleading when the object of interest is the complete daily profile. Indeed, even if the 24 coverage events were independent, the probability that the entire trajectory lies simultaneously inside 24 separate 90% intervals would be only $0.9^{24}=8\%$.

In practice, forecast errors across neighbouring hours are also dependent: underestimating demand at 8 a.m. often provides information about the error at 9 a.m. or 10 a.m.

The project therefore asks:

> "How can dependence between several predicted quantities be exploited to construct joint prediction regions that contain the complete demand trajectory with the desired probability, without becoming unnecessarily large?"

Students will begin with marginal prediction intervals and simple joint regions, then study approaches that explicitly use correlations between forecast errors. The research-oriented extension will consider more flexible multivariate methods based on geometric quantiles and optimal transport.

The project will also study how these methods behave as the dimension increases, for example by representing a day using 2, 4, 6, 12 or 24 time blocks. Controlled simulations will complement the electricity application and help identify which dependence structures more sophisticated methods actually handle advantageously.

The main comparison will be between "joint coverage" (how often the complete future demand vector is contained in the

predicted region) and the "size of the region". A useful method should achieve the desired coverage while excluding as many implausible demand trajectories as possible.

Données du sujet :

The project will use the public "RTE éCO2mix" dataset for France. RTE provides realized national electricity consumption, along with the "J-1 forecast" produced the previous day, at a fine temporal resolution. Historical national observations are available from 2012.

Students will construct daily vectors of realized and predicted consumption and analyze the corresponding forecast errors. The data will be aggregated at several temporal resolutions in order to vary the dimension of the prediction problem.

Modèles envisagés du sujet :

Standard prediction models

Bibliographie du sujet :

- RTE. éCO2mix ? consommation, production et prévisions d'électricité.
- Chaudhuri, P. (1996). On a Geometric Notion of Quantiles for Multivariate Data. JASA.
- Hallin, M., del Barrio, E., Cuesta-Albertos, J. A., & Matrán, C. (2021). Distribution and Quantile Functions, Ranks and Signs in Dimension d: A Measure Transportation Approach. Annals of Statistics.
- Xu, C., Jiang, H., & Xie, Y. (2024). Conformal Prediction for Multi-Dimensional Time Series by Ellipsoidal Sets. ICML.
- Thurin, G., Nadjahi, K., & Boyer, C. (2025). Optimal Transport-Based Conformal Prediction. ICML.
- Peyré, G., & Cuturi, M. (2019). Computational Optimal Transport.

Notation du sujet :

Commentaires :